CM2017 P4 Project Plan – MoveSync – 2026/03/24

Project Overview

MoveSync is a browser-based motion analysis tool built on HTML/CSS/JavaScript that combines inertial measurement unit (IMU) data with synchronised video to visualise and quantify athletic movement. The platform is designed for sport scientists, coaches, and students who need flexible, convenient access to biomechanical insights without requiring specialist hardware or installed software.

Team Members and Roles

Member	Role	Primary Responsibilities
Alex Jonsson	*Developer*	*Signal Processing/Metrics*
Jack Hutton	*Developer*	*IMU Data Processing/Metrics*
Sahand Khodabandehloo	*Developer*	*Frontend Development*
Jonas Willén	*Supervisor*	*Guidance and Expertise*

Project Objectives

Short Term Goals (Weeks 1-2)

Before introducing new functionality, it is important to strengthen the current system and address any weaknesses in the existing implementation. Therefore, the initial focus during the first two weeks of PCC3 will be on refining the application developed in PCC2 to ensure it is stable, maintainable, and ready for the improvements planned. This phase will concentrate on the following areas:

- **Bug fixes and stabilization**

Establish a more reliable foundation and resolve known defects across all existing features before beginning new development. As the application continues to grow across different modalities (e.g., new functionalities and enhancements in the *Library* and *Session Viewer*), the likelihood of new bugs also increases. Several issues have already been identified in the current application; however, there may be additional undiscovered defects. During earlier development, bugs, problematic code, and general stability improvements were addressed continuously. This same iterative approach of identifying, addressing, and resolving issues will continue throughout PCC3.

- **Layout simplification**

Improve the usability and clarity of the *Session Viewer* presentation by simplifying the layout. This will involve consolidating panels where appropriate and introducing hide/show functionality to reduce visual clutter. The aim is to create a cleaner and more intuitive interface that allows users to focus on the most relevant information while still having access to additional details when needed.

- **GitHub documentation**

Update the public GitHub repository documentation, including the README and relevant technical specification documents. This will ensure that the project is clearly documented, easier to understand for new contributors, and better aligned with the current state of the application. Improved documentation will also support maintainability and provide a clearer record of design decisions and system structure.

- **Improve timestamp functionality**

Extend the annotation and timestamp functionality to better support future session comparison features. This includes improving how timestamps are created, accessed, and managed within the system, so they can serve as a stronger basis for comparing sessions in a structured and meaningful way (in the *Compare Sessions* modality).

- **Fix outdated or placeholder code**

Review the existing codebase to remove unnecessary or obsolete code and replace temporary placeholder implementations, (e.g., the sports metrics list), with more complete and appropriate solutions. This will help reduce technical debt, improve code clarity, and make the system easier to maintain and extend in later stages of development.

Long Term Goals (Weeks 3-9)

The longer-term phase of PCC3 will focus on introducing more advanced functionality and improving the analytical value of the application. Once the system has been stabilised and refined during the initial weeks, development can shift towards features that expand the application's usefulness for comparison, coaching, education, and multi-sensor analysis. This phase will include the following goals:

- **Session comparison**

Develop a side-by-side comparison view for different sessions and timestamps, allowing synchronised data and key metrics to be analysed together. This feature will make it easier to identify differences in performance across sessions, support more meaningful review of recorded data, and provide a stronger basis for both coaching and evaluation.

- **Sports metrics accuracy**

Benchmark and improve the accuracy of speed, distance, and other important sports-related metrics. This will involve tuning ZUPT thresholds, adjusting filter parameters, and testing multiple signal-processing methods to compare performance and reliability. The aim is to ensure that the reported metrics are both consistent and as accurate as possible across different scenarios.

- **Educational lab**

Develop the application so that it can be used as part of a structured educational lab experience, as proposed by the supervisor, Jonas Willén. This includes enabling modifiable signal-processing parameters so that users can interact with and explore the effects of different processing choices. This would make the software not only a practical analysis tool, but also a useful learning platform for demonstrating relevant technical concepts.

- **Multiple IMUs support**

Reintroduce support for multiple IMUs to enable more advanced and comprehensive analysis. Supporting multiple sensors would allow the system to capture a broader range of motion data, improve analytical depth, and create opportunities for more detailed comparisons between different body positions or movement patterns.

- **Additional coaching tools**

Add further coaching-oriented features (e.g., video region marking and annotation export) that improve review and feedback workflows. These additions are intended to make the application more useful in practical coaching settings by supporting clearer communication, easier session review, and more effective performance analysis.

Development Approach

The development plan is detailed in the Gantt chart appended to the end of this document. It follows a quality-first principle through an initial Phase 1 focused on the short-term goals of ensuring the current application is stable and well documented. Following this, Phase 2 will focus on the long-term goals prioritising Session Comparison and Metric Accuracy.

Work on the sports metrics will be first researched then implemented with multiple processing approaches compared, preserved and documented to support the long-term educational aid goal. All work will be completed initially in a private shared GitHub, with regular updates being pushed to the publicly available repository.

The team will meet weekly with the project supervisor Jonas Willén to review progress, discuss challenges, and adjust priorities as needed. Occasional meetings with external experts will hopefully be arranged to provide additional guidance to the project.

Key Dates

Milestone	Date
Project Plan Submission	*26/03/29*
JML Memoir Submission (Individual)	*26/04/26*
Preliminary Presentation	*26/04/27*
Preliminary Report Submission	*26/05/03*
Preliminary Peer Reviews	*26/05/10*
Final Report Submission	*26/05/24*
Final Presentation	*26/05/26*

IMU Video Project

Team Members	Jack, Sahand, Alex
Supevisor	Jonas Willén
START	**23/03/2026**
DISPLAY WEEK	**1**
END	**28/05/2026**

Task	ASSIGNED TO	START DATE	DAYS	END DATE
Phase 1 - Short Term		**23/03/2026**	**4**	**5/04/2026**
Layout Simplification	All	23/03/2026	7	29/03/2026
Current Feature Fixes	All	23/03/2026	7	29/03/2026
Update GitHub Docs	All	30/03/2026	2	31/03/2026
Improve Timestamps	All	1/04/2026	5	5/04/2026
Fix Sports Metrics Page	All	1/04/2026	5	5/04/2026
Phase 2 - Long Term		**30/03/2026**		**17/05/2026**
Session Comparison	All	30/03/2026	21	19/04/2026
Sport Metric Accuracy	All	6/04/2026	35	10/05/2026
Educational Lab	All	20/04/2026	28	17/05/2026
Multiple IMUS	All	27/04/2026	21	17/05/2026
Additional Coaching Tools	All	4/05/2026	14	17/05/2026
Assignments		**23/03/2026**		**26/05/2026**
Project Plan	All	23/03/2026	7	29/03/2026
JML Memoir	Individual	20/04/2026	7	26/04/2026
Prelimiary Presentation	All	22/04/2026	6	27/04/2026
Preliminary Report	All	27/04/2026	7	3/05/2026
Final Report	All	11/05/2026	14	24/05/2026
Final Presentation	All	20/05/2026	7	26/05/2026

